

Impact Of Membrane Polyunsaturated Fatty Acids on D2R Heteromerization and Signaling

Garima C.N Thakur, Nejma Rabhi, Mathias Lescuyer, Thierry Durroux, Sandrine Villette, Marie-Lise Jobin, Isabel Alves, Estelle Rascol

Institut de Chimie et de Biologie des Membranes et des nano-objets, Université de Bordeaux, CNRS, UMR 5248, 33600 Pessac, France

Keywords: GPCRs, plasma membrane, lipids, oligomerization, signaling

ABSTRACT

Schizophrenia is a severe neuropsychiatric disorder affecting 24 million people worldwide, characterized by immense clinical and neurobiological heterogeneity [1]. Traditional antipsychotic strategies primarily target the monomeric Dopamine D2 Receptor (D2R), but these approaches often fall short by ignoring physiological receptor cross-talk [3]. In the central nervous system, D2R physically interacts with the Cannabinoid 1 Receptor (CB1R) to form a distinct pharmacological entity with fundamentally reprogrammed G-protein coupling profiles. While systemic lipid disturbances are associated with the pathology [2] and recent data suggest allosteric modulation of D2R activity by the local membrane lipid environment [4,5], the precise molecular architecture of this heteromeric complex and how its physical interface drives unique intracellular signaling dynamics remain entirely unknown.

In this context, this project investigates how polyunsaturated fatty acids (PUFAs), modulate D2R-CB1R heteromerization and its subsequent downstream signalling. This homodimer target is justified by preliminary time-resolved fluorescence resonance energy transfer (TR-FRET) data demonstrating interaction between both receptors. While each receptor individually couples to the inhibitory $G_{i/o}$ pathway, their co-stimulation in co-expressing cells is reported to lead to an unexpected increase in cAMP production, attributed to a functional switch to G_s activation that remains to be proved. To dissect this mechanism, we first characterize D2R and CB1R lateral diffusion and co-localization at the plasma membrane using Total Internal Reflection Fluorescence (TIRF) microscopy combined with Single Particle Tracking (SPT) analysis. Second, using a combination of energy transfer biosensors we are mapping the signaling cross-talk in *cellulo*. In parallel, both receptors will be purified to characterize their direct physical interaction *in vitro*, effectively decoupling core receptor-receptor coupling from confounding downstream cellular events.

REFERENCES

- 1 Wang, D. et al. Characterising phospholipids and free fatty acids in patients with schizophrenia : A case-control study Characterising phospholipids and free fatty acids in patients with. *World J Biol Psychiatry*. 2020;0(0):1-14. doi:10.1080/15622975.2020.1769188
- 2 Hoen, W. P. et al.Red blood cell polyunsaturated fatty acids measured in red blood cells and schizophrenia: a meta-analysis. *Psychiatry Res* 2013, 207 (1-2), 1-12. DOI: 10.1016/j.psychres.2012.09.041
- 3 Fuxe, K. et al. Dopamine D2 heteroreceptor complexes and their receptor-receptor interactions in ventral striatum: novel targets for antipsychotic drugs. *Prog Brain Res* 2014, 211, 113-139. DOI: 10.1016/B978-0-444-63425-2.00005-2
- 4 Sposini, S. et al. Membrane lipid poly-unsaturation selectively affects dopamine D2 receptor internalization. Preprint bioRxiv 2026 <https://doi.org/10.1101/2023.12.14.571632>
- 5 Baccouch, R. et al. The role of the lipid environment in the activity of G protein coupled receptors. *Bio-phys Chem* 2022, 285, 106794. DOI: 10.1016/j.bpc.2022.106794

Funding : Fondation La maison de la chimie, ANR SCHIZOLIP and STS department of CBMN